Natural Language Processing Basics

Original educational benchmark document for testing document retrieval and grounded question answering.

1. Text Preparation

Natural language processing (NLP) is a field concerned with enabling computers to work with human language. NLP applications include text classification, information extraction, search, translation, and question answering.

Tokenization divides text into smaller units called tokens. Depending on the method, a token may be a word, a subword unit, or a character. Tokenization is an important step because models operate on token sequences rather than raw sentences.

Normalization converts text into a more consistent form. Common operations include lowercasing and standardizing whitespace. Stemming reduces words to approximate stems, while lemmatization maps words to dictionary base forms using linguistic information.

Stop words are frequent words that some text-processing pipelines remove when they contribute little to a particular task. Stop-word removal is not always appropriate; words such as 'not' can be essential to meaning.

2. Representing Text

A bag-of-words representation describes a document using counts or weights of terms while ignoring most word-order information. It is simple and useful for some tasks, but it does not directly represent the sequence of words.

Term frequency measures how often a term occurs in a document. Inverse document frequency reduces the weight of terms that appear in many documents. TF-IDF combines these ideas to give higher weight to terms that are frequent in a document but less common across the collection.

An embedding is a dense numerical vector intended to represent a word, sentence, or document. Embeddings can place texts with related meanings near one another in vector space.

Cosine similarity compares the direction of two vectors. It is commonly used to estimate similarity between text embeddings, with larger values indicating more similar directions.

3. Language Models and Retrieval

A language model estimates patterns in sequences of tokens and can generate text based on context. A transformer is a neural-network architecture that uses attention mechanisms to process relationships between tokens.

Retrieval-augmented generation (RAG) combines information retrieval with text generation. A retrieval component selects relevant passages from a collection, and a language model uses those passages as context when producing an answer.

A chunk is a smaller segment of a source document used for indexing and retrieval. Chunk size and overlap affect how much context is preserved and how passages are retrieved.

Grounded question answering should distinguish information supported by retrieved passages from information that is not present in the supplied context. Citations should point to the source passages used to support an answer.